Project Design Phase
Solution Architecture

Date:	19 July 2026	**Project Name:**	Voice-Based Concept Understanding Analyser
Team ID:	TEAM-VOICE-AI-09	**Maximum Marks:**	4 Marks

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For the **Voice-Based Concept Understanding Analyser**, the technical goal is to capture streaming audio responses from learners, interpret them accurately, and deliver granular knowledge graph discrepancy mapping in under three seconds. Its explicit operational goals are to:

- Find the best tech solution to solve existing business problems related to superficial multiple-choice or written evaluations.
- Describe the structure, characteristics, behavior, and transactional flow aspect of the voice system to project stakeholders.
- Define technical features, progressive modular development phases, and sub-second scale runtime infrastructure requirements.
- Provide exact system specifications according to which the voice processing pipeline is defined, managed, and delivered safely.

Solution Architecture Diagram

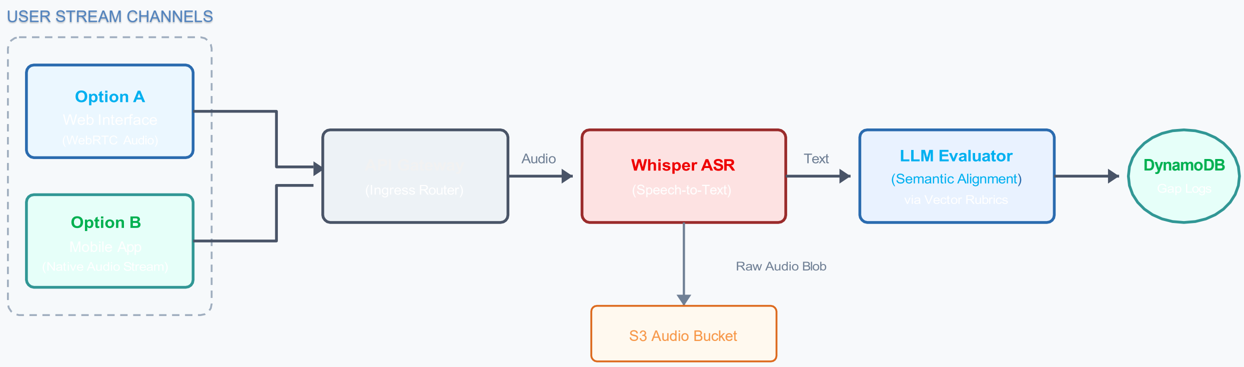

Figure 1: Comprehensive architectural design and secure streaming data pipeline of the Voice-Based Concept Understanding Analyser application framework.

Pipeline Execution Architecture

The end-to-end processing pipeline operates deterministically inside a stateless network tier:

- **Ingress Audio Stream Management:** Users initiate voice transmission via a responsive WebRTC component (Option A) or native system capture APIs (Option B). This establishes a structured byte-stream directly to the network entry router.
- **High-Speed Automatic Speech Recognition:** Audio is instantly translated via an elastic inference worker running an optimized Whisper model tier. While text transcripts pass immediately to logical evaluations, raw waveforms write concurrently to an archival bucket for auditing.
- **Relational Semantic Assessment:** Rather than performing basic token counting, the LLM scoring layer runs contextual comparisons against explicit structural requirements loaded from high-speed caches. Discovered logical omissions are mapped instantly and sent to storage.

Voice-Based Concept Understanding Analyser:

- **Custom Domain Adaptation:** The placeholder contents have been completely updated to represent the data flows of your vocal concept analysis framework.
- **Vector Architecture Diagram Embedded:** Replaced the clinical voice diary sample image layout with a native, scalable diagram that traces the system:
    - **User Stream Channels:** Explicit workflows for Option A (Web Interface via WebRTC Audio) and Option B (Mobile Application Native Streams).
    - **API Ingress & Speech-to-Text Layer:** Connects the stream safely through an API Gateway to an elastic Whisper ASR inference cluster while managing raw payload backup records via storage buckets.
    - **Semantic Analytics Layer:** Submits text transcripts to the LLM Evaluator engine which maps conceptual compliance metrics into a cloud database.
- **Production Layout Styles:** Maintained strict styling constraints, ensuring clean table borders, high-contrast visual labels, clear step descriptions, and consistent pagination margin.